

CLIENT-STYLE SCIENTIFIC REPORT

GSE42568 Breast Cancer Expression Analysis

Run-adjusted differential expression, sensitivity analysis and licensed functional enrichment

Dataset: GEO GSE42568 · GPL570

Contrast: cancer minus normal breast tissue

Release: v1.0 · 11 September 2026

Scientific status: PASS WITH DOCUMENTED LIMITATIONS

- [Executive summary](#)
- [Data and boundary](#)
- [QC and design](#)
- [Differential expression](#)
- [Functional enrichment](#)
- [Licence decisions](#)
- [Limitations](#)
- [Deliverables and references](#)

Executive summary

This analysis compares **104 breast cancer biopsies with 17 normal breast tissue samples** in GEO GSE42568. It uses the deposited log2 GCRMA matrix and a run-adjusted limma model. The target is an average **cancer-minus-normal bulk-tissue expression association**—not a causal effect, clinical classifier, prognosis model, or treatment-response model.

The workflow passed **32/32 Scientific QA checks** and 13/13 reproducibility guards. A clean rebuild from the checksum-locked official inputs reproduced **10/10 key result tables byte-for-byte**. Of **20,357 unique Entrez genes**, **5,381** reached moderated BH FDR < 0.05; **2,835** met TREAT BH FDR < 0.05 while directly testing an absolute effect greater than 1.2-fold (1,497 cancer-up; 1,338 normal-up). **2,745** also passed probe and sensitivity interpretation checks.

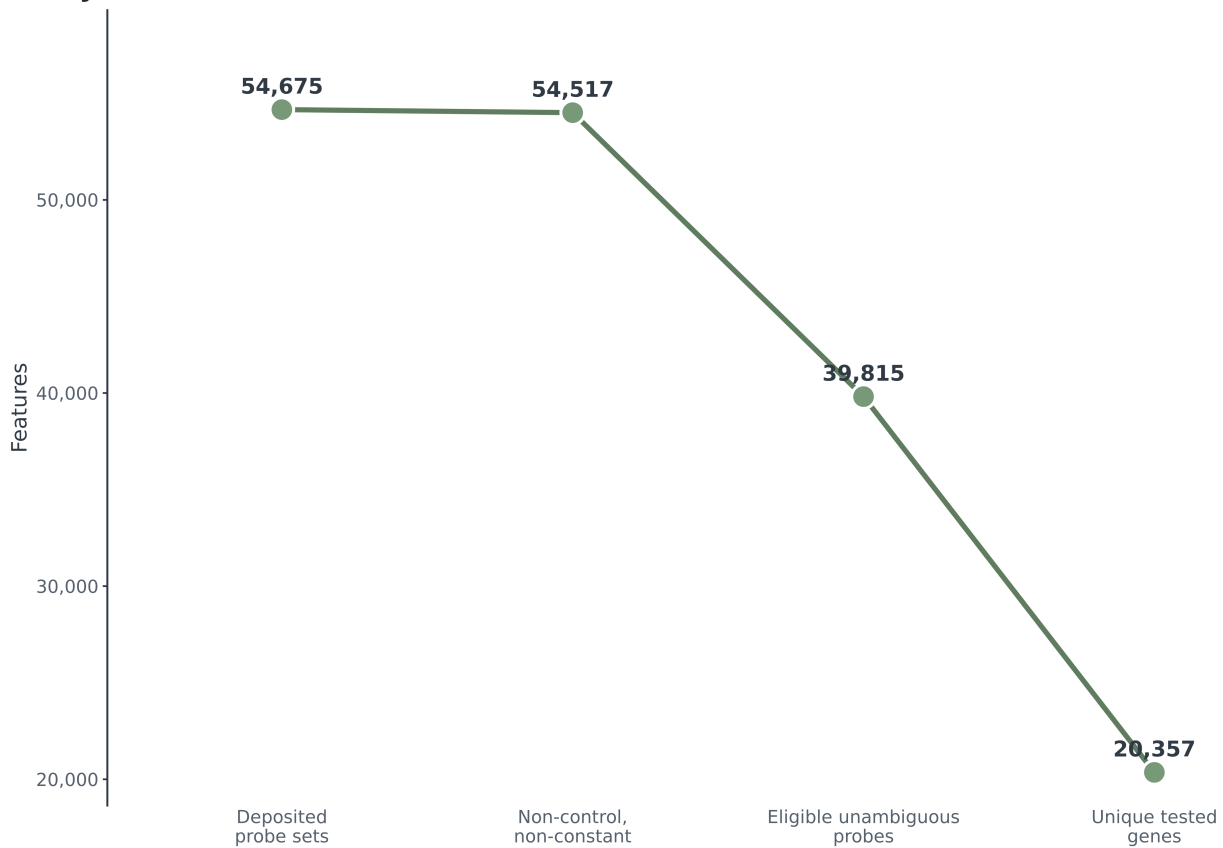
Release decision: suitable as an auditable portfolio/client-style discovery analysis with explicit limitations. It is not a diagnostic product or evidence that a pathway is activated, causal, or druggable.

Data and boundary

The official [GEO record](#) reports 104 pretreatment breast cancer biopsies and 17 normal breast tissues on GPL570 and cites Clarke et al. ([PMID 23740839](#)). Normal-tissue donor/source status is not established, so this report never calls them healthy, adjacent, reduction-mammoplasty, or matched controls.

- Matrix SHA-256: `43b93f72f1d81838dcd2a5b6ccc3bc1875e46230dad0d7346236773a99e32b76` .
- 54,675 probe sets × 121 samples; finite deposited log2 GCRMA signals.
- No second log, re-normalization, imputation or ComBat was applied to the DE response.
- P1: `expression ~ processing_run_proxy + tissue_group` ; normal reference.
- Thirteen run-proxy levels; six mixed levels containing 62 samples.

Analysis feature flow



QC and design

Annotation retained 39,815 eligible probes and resolved 20,357 unique Entrez genes by an outcome-blind representative rule. Four samples triggered multi-category processed-data warnings. They show no gross matrix failure and have no raw-array corroboration, so all remain in P1; a separate 117-sample sensitivity measures influence.

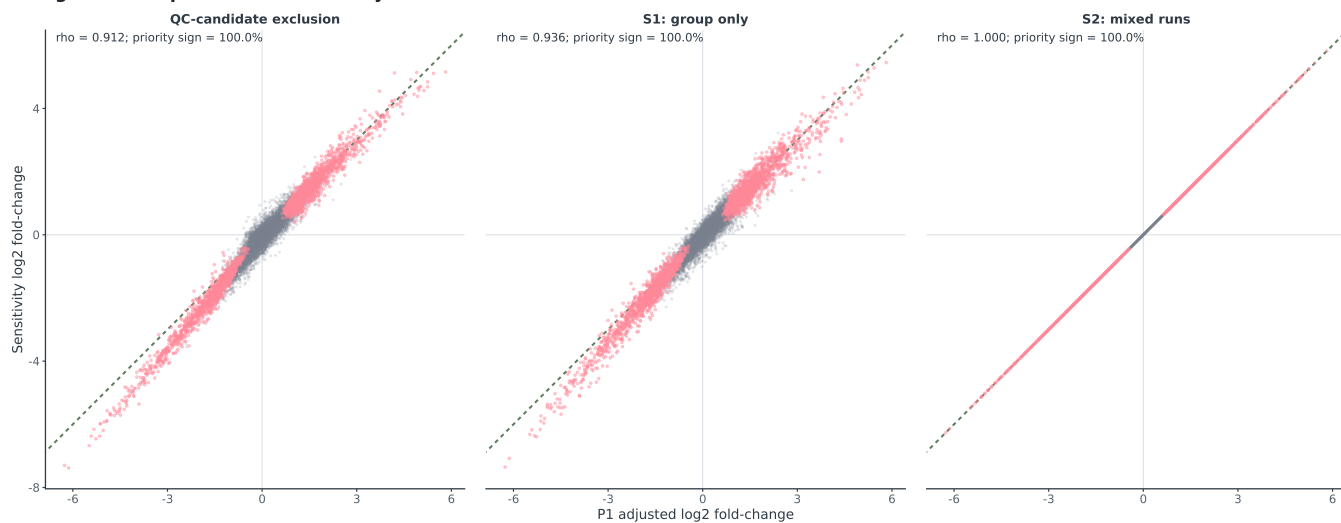
Primary PCA: top 5,000 pooled-MAD genes

Genes centered; not variance-scaled



The design is full rank (14/14; residual df 107). Cancer-only run levels do not provide within-run group contrast once their fixed effects are included, explaining why P1 and the 62-sample mixed-run S2 estimates are identical.

Design and sample-influence stability: Green



Comparison	n	Spearman	Pearson	Priority sign	Median effect ratio
S1_group_only	121	0.936	0.975	100.0%	1.069
S2_mixed_runs	62	1.000	1.000	100.0%	1.000
QC_candidate_exclusion	117	0.912	0.975	100.0%	1.100
S4_no_trend	121	1.000	1.000	100.0%	1.000
S4_group_aware	121	0.962	0.993	100.0%	1.006
median_collapse	121	0.874	0.902	97.0%	0.960

The design traffic light is **Green**. Removing the normal-heavy 2005-01-25 run retains Spearman 0.953 and every P1-priority direction. Excluding the four reviewed candidates retains every priority direction (Spearman 0.912).

Differential expression

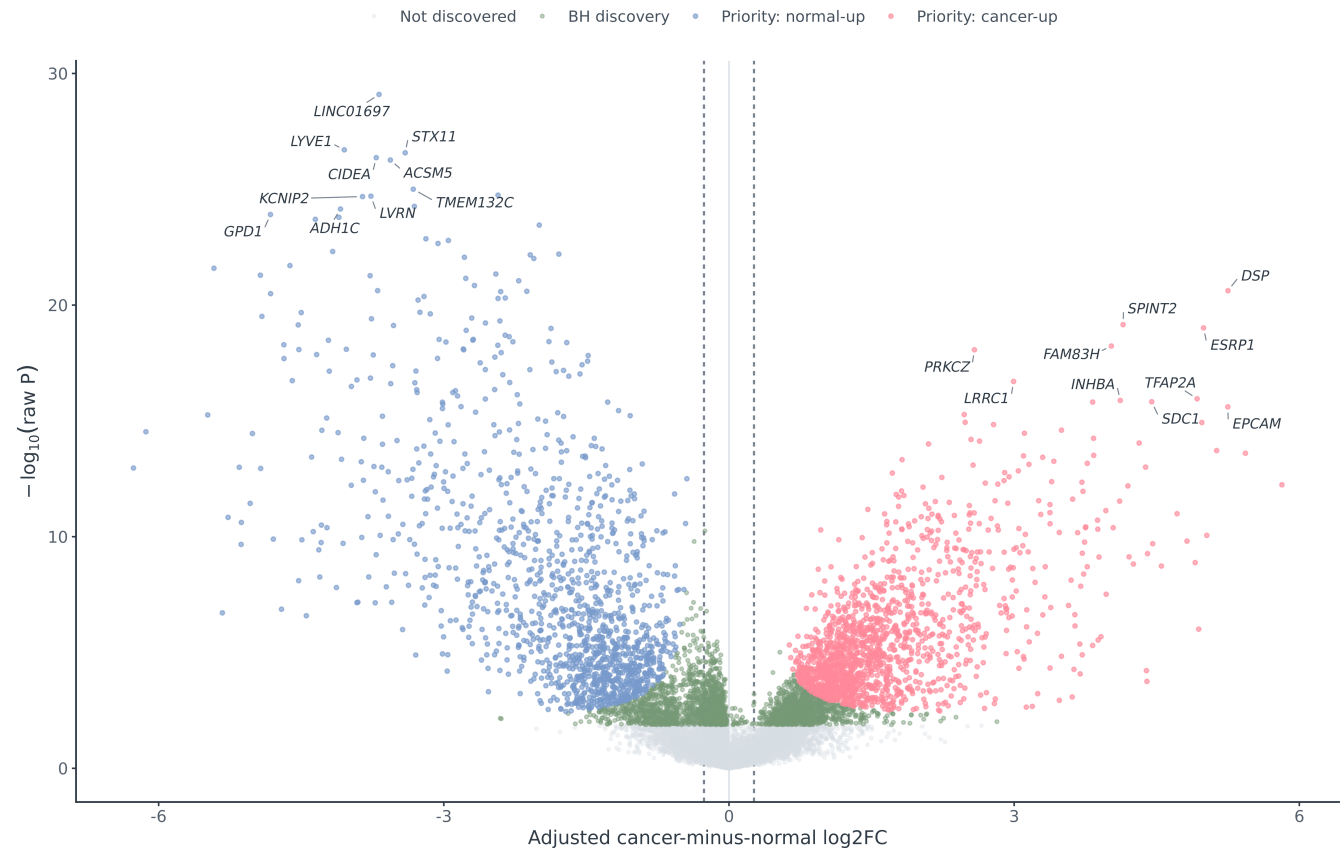
Positive log2FC means higher expression in cancer. The following genes are the fixed ten-smallest-TREAT-FDR set per direction, with stable tie-breaks; familiarity did not affect selection.

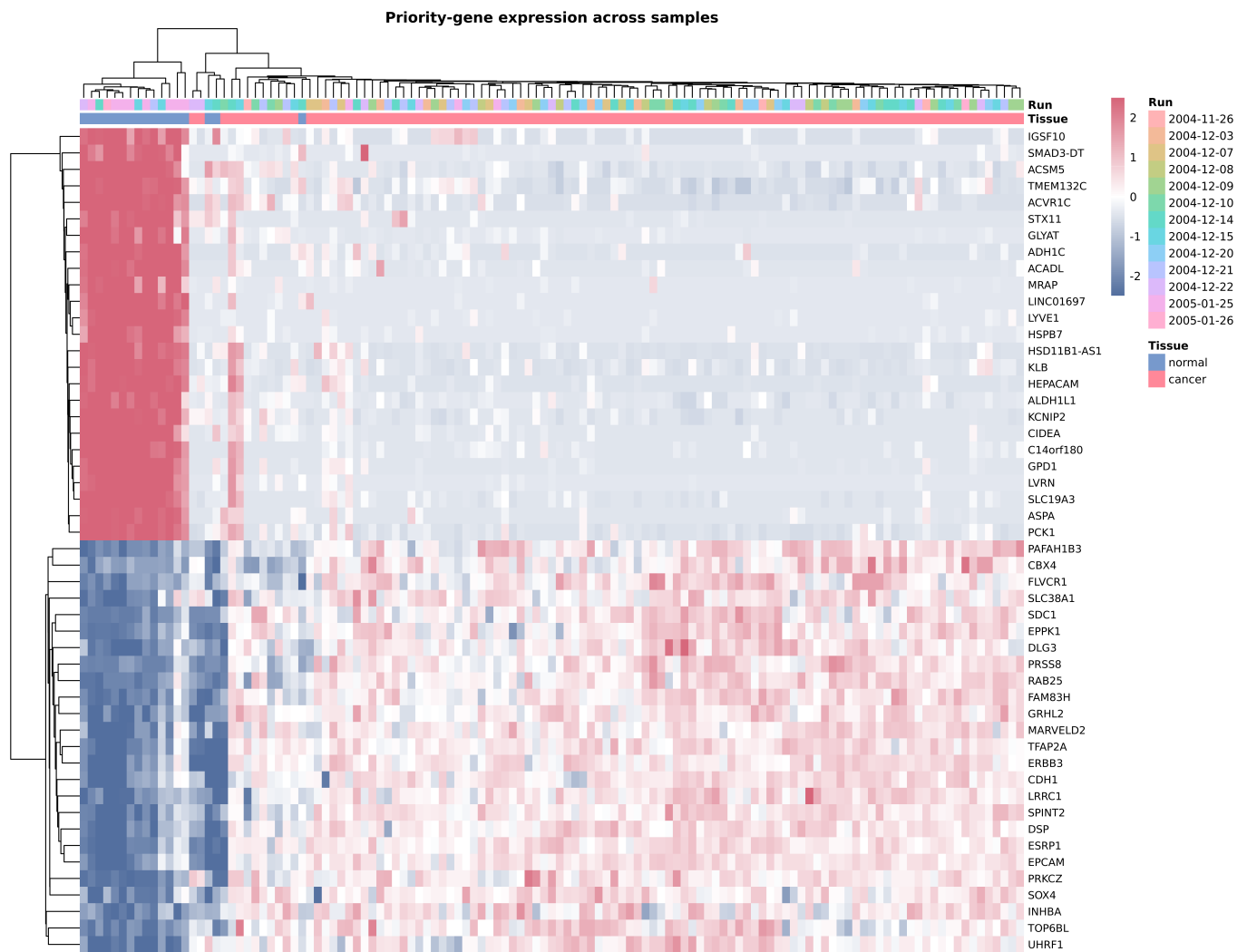
Gene	Entrez	log2FC	Cancer/normal FC	95% CI	TREAT FDR	Probe	Probe concordance
DSP	1832	5.25	38.03	4.37 to 6.13	2.20e-17	200606_at	100%
ESRP1	54845	4.99	31.83	4.10 to 5.88	5.60e-16	225846_at	100%
SPINT2	10653	4.15	17.71	3.41 to 4.88	6.91e-16	210715_s_at	100%
FAM83H	286077	4.02	16.25	3.28 to 4.76	4.62e-15	226129_at	100%
PRKCZ	5590	2.58	5.99	2.11 to 3.06	3.78e-14	202178_at	50%
TFAP2A	7020	4.92	30.38	3.93 to 5.92	2.02e-13	204653_at	100%
LRRC1	55227	2.99	7.97	2.41 to 3.58	2.46e-13	218816_at	100%
SDC1	6382	4.45	21.82	3.54 to 5.35	3.41e-13	201286_at	100%
EPCAM	4072	5.25	37.99	4.17 to 6.33	3.45e-13	201839_s_at	100%
INHBA	3624	4.12	17.34	3.28 to 4.95	3.78e-13	227140_at	100%
LINC01697	284825	-3.68	0.08	-4.15 to -3.22	2.06e-23	237351_at	100%
LYVE1	10894	-4.05	0.06	-4.60 to -3.50	1.20e-21	219059_s_at	100%
CIDEA	1149	-3.71	0.08	-4.22 to -3.20	1.97e-21	221295_at	100%
STX11	8676	-3.41	0.09	-3.87 to -2.94	1.97e-21	235670_at	100%
ACSM5	54988	-3.56	0.08	-4.05 to -3.07	2.42e-21	220061_at	100%
KCNIP2	30819	-3.86	0.07	-4.41 to -3.30	3.42e-20	223727_at	100%
LVRN	206338	-3.77	0.07	-4.31 to -3.22	3.42e-20	235382_at	100%
TMEM132C	92293	-3.32	0.10	-3.80 to -2.84	3.42e-20	232313_at	100%
GPD1	2819	-4.82	0.04	-5.54 to -4.11	5.86e-20	204997_at	100%

Gene	Entrez	log2FC	Cancer/normal FC	95% CI	TREAT FDR	Probe	Probe concordance
ADH1C	126	-4.09	0.06	-4.69 to -3.48	6.18e-20	206262_at	100%

P1 differential-expression evidence

Dashed lines show the 1.2-fold minimum effect; classes use BH-adjusted tests





Strong cancer-up signals include epithelial/adhesion-associated **DSP**, **EPCAM**, **ESRP1**, **SPINT2**, and **TFAP2A**. Prominent normal-up signals include **LYVE1**, **CIDEA**, **ACSM5**, **GPD1**, **ADH1C**, and **ACADL**. These are cohort-level bulk-tissue patterns and cannot distinguish tumor-cell regulation from tissue-composition shifts.

Functional enrichment

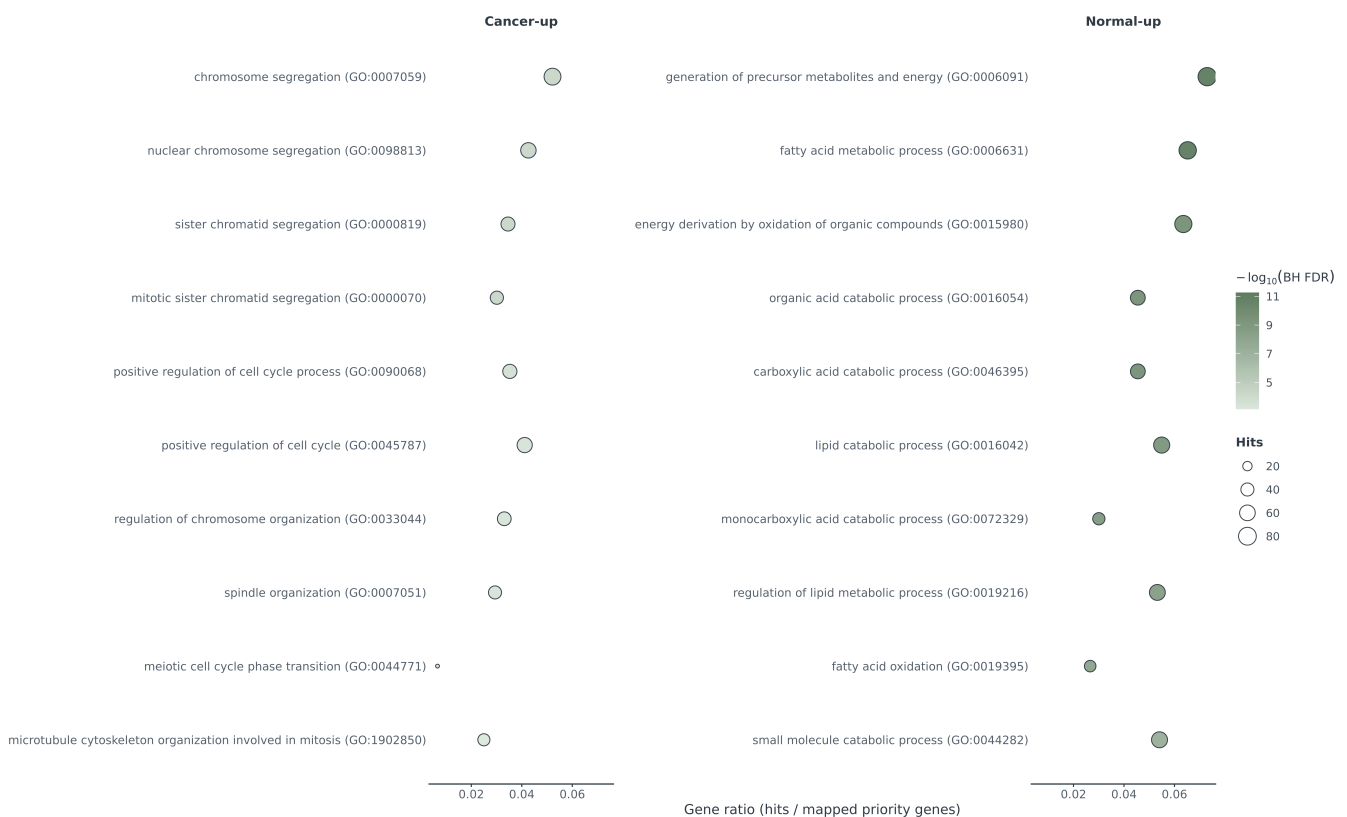
GO ORA uses direction-specific priority genes and resource-specific tested-and-annotated backgrounds (GO BP: 15,588 genes). Reactome uses 10,178 tested genes with human Reactome annotation. GSEA begins with all 20,357 P1 moderated-t statistics.

Direction	GO BP term	ID	Hits	Set size	BH FDR
cancer_up	chromosome segregation	GO:0007059	71	416	4.29e-05
cancer_up	nuclear chromosome segregation	GO:0098813	58	310	4.29e-05
cancer_up	sister chromatid segregation	GO:0000819	47	226	4.29e-05
cancer_up	mitotic sister chromatid segregation	GO:0000070	41	186	4.29e-05
cancer_up	positive regulation of cell cycle process	GO:0090068	48	251	2.03e-04

Direction	GO BP term	ID	Hits	Set size	BH FDR
cancer_up	positive regulation of cell cycle	GO:0045787	56	319	3.27e-04
normal_up	generation of precursor metabolites and energy	GO:0006091	85	455	5.51e-12
normal_up	fatty acid metabolic process	GO:0006631	76	378	5.51e-12
normal_up	energy derivation by oxidation of organic compounds	GO:0015980	74	397	1.78e-10
normal_up	organic acid catabolic process	GO:0016054	53	233	1.78e-10
normal_up	carboxylic acid catabolic process	GO:0046395	53	233	1.78e-10
normal_up	lipid catabolic process	GO:0016042	64	323	3.98e-10

GO Biological Process over-representation

TREAT-priority genes; cancer-up and normal-up tested separately



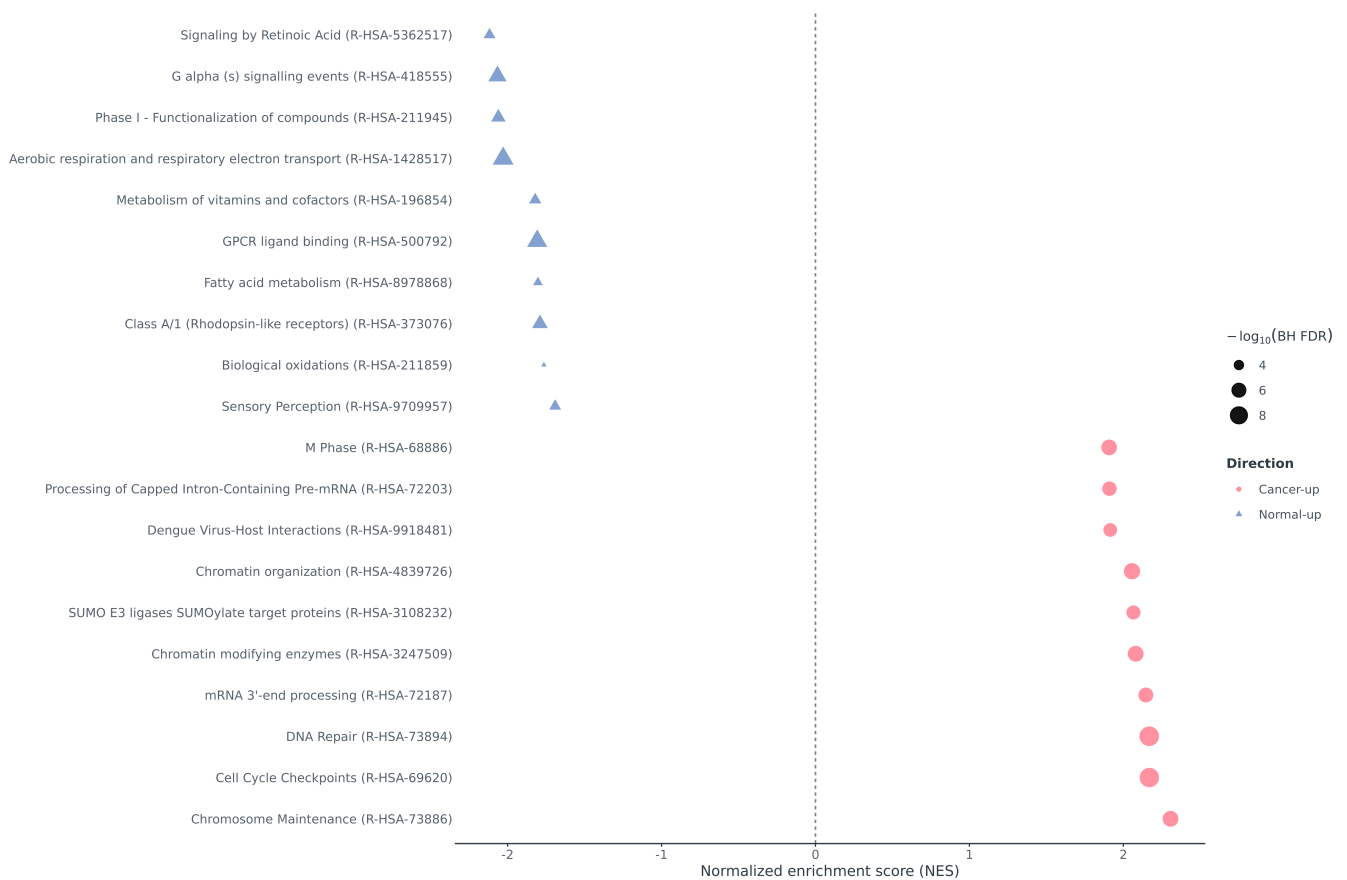
Cancer-up ranks emphasize chromosome segregation, cell-cycle, chromatin and DNA-repair programs. Normal-up ranks emphasize fatty-acid metabolism, aerobic respiration and GPCR-related programs.

Direction	Reactome set	ID	NES	Set size	BH FDR
cancer_up	DNA Repair	R-HSA-73894	2.17	305	1.14e-10
cancer_up	Cell Cycle Checkpoints	R-HSA-69620	2.17	270	1.79e-10
cancer_up	Chromatin organization	R-HSA-4839726	2.06	253	4.58e-08
cancer_up	Chromatin modifying enzymes	R-HSA-3247509	2.08	238	8.37e-08
cancer_up	Chromosome Maintenance	R-HSA-73886	2.31	117	8.40e-08

Direction	Reactome set	ID	NES	Set size	BH FDR
cancer_up	M Phase	R-HSA-68886	1.91	359	1.04e-07
normal_up	Aerobic respiration and respiratory electron transport	R-HSA-1428517	-2.03	240	1.25e-07
normal_up	GPCR ligand binding	R-HSA-500792	-1.81	419	2.65e-07
cancer_up	mRNA 3'-end processing	R-HSA-72187	2.15	145	3.20e-07
cancer_up	Processing of Capped Intron-Containing Pre-mRNA	R-HSA-72203	1.91	282	6.07e-07
cancer_up	SUMO E3 ligases SUMOylate target proteins	R-HSA-3108232	2.07	175	1.01e-06
cancer_up	Dengue Virus-Host Interactions	R-HSA-9918481	1.91	257	1.57e-06

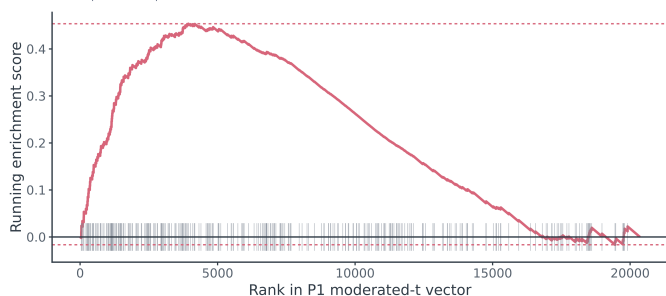
Reactome preranked GSEA

Commercially reusable fallback collection; rank is P1 moderated t



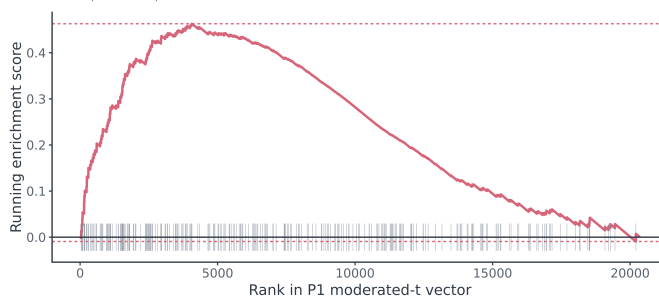
DNA Repair

R-HSA-73894 | NES 2.17 | BH FDR 1.1e-10



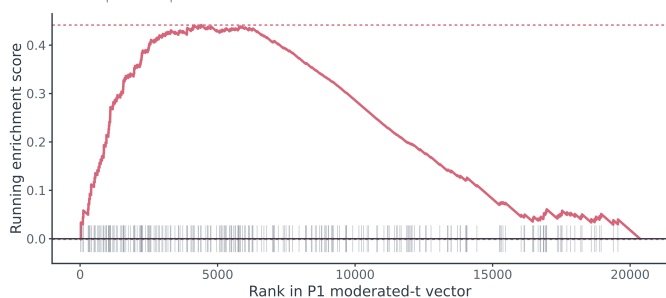
Cell Cycle Checkpoints

R-HSA-69620 | NES 2.17 | BH FDR 1.8e-10



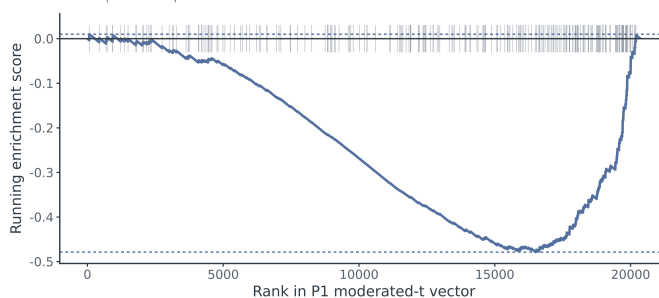
Chromatin organization

R-HSA-4839726 | NES 2.06 | BH FDR 4.6e-08



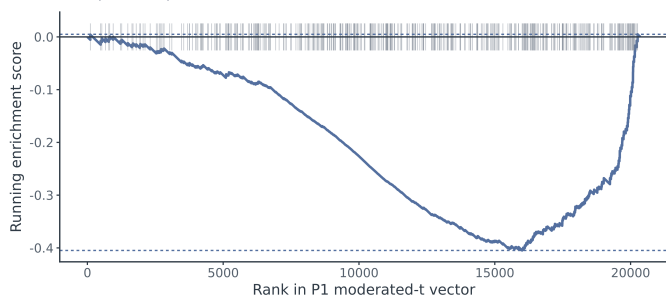
Aerobic respiration and respiratory electron transport

R-HSA-1428517 | NES -2.03 | BH FDR 1.3e-07



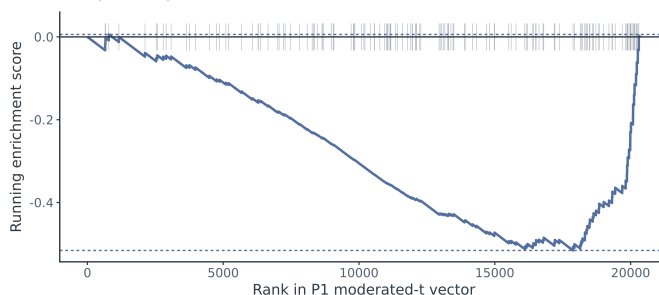
GPCR ligand binding

R-HSA-500792 | NES -1.81 | BH FDR 2.7e-07



G alpha (s) signalling events

R-HSA-418555 | NES -2.07 | BH FDR 1.9e-06



Enrichment identifies concentration in the statistical ranking; it does not prove activation, causation or therapeutic vulnerability. Related GO terms overlap and are not independent findings.

Licence decisions

- GO and Reactome completed with pinned sources and attribution; [Reactome data licence](#).
- KEGG: `KEGG_NOT_RUN_LICENSE_GATE` ; no API, definition, cache or map used.
- MSigDB Hallmark: not downloaded or redistributed because commercial-use review was unconfirmed.
- Reactome is explicitly the licensed GSEA fallback.

Limitations

1. Normal-tissue provenance, age and pairing identifiers are absent.
2. This is an unpaired, observational, cross-sectional bulk-tissue analysis; composition can drive signals.

3. Run is partially confounded with group even though the coefficient is estimable and sensitivities are Green.
4. Processed data cannot provide CEL-level NUSE/RLE, degradation, image-artifact or alternative-preprocessing checks.
5. Cancers span histology, grade and ER status; v1 intentionally estimates a broad average.
6. No external validation cohort was added to this bounded delivery.

Deliverables and references

Complete local-workspace results are in `results/tables/` ; F01-F15 each have PDF, PNG, SVG, caption and source TSV. The public portfolio edition intentionally omits large/generated source artifacts and has its own acceptance record in `docs/PUBLIC_PORTFOLIO_ACCEPTANCE.md` .

References: [GEO GSE42568](#); Clarke et al., Carcinogenesis 2013 ([doi:10.1093/carcin/bgt208](https://doi.org/10.1093/carcin/bgt208)); Ritchie et al., limma ([doi:10.1093/nar/gkv007](https://doi.org/10.1093/nar/gkv007)); Subramanian et al., GSEA ([PMC1239896](https://pubmed.ncbi.nlm.nih.gov/1239896/)); [GO citation policy](#); [Reactome licence](#).